CLINICAL · EAR

Hearing rehabilitation: aids, bone devices and implants

Hearing rehabilitation means matching the device to the loss. A hearing aid helps whatever amplification can help; a bone-conduction device helps when the canal or middle ear is in the way; a cochlear implant takes over when the cochlea can no longer do the job; and a brainstem implant is kept for the patient who has no cochlear nerve.

AT A GLANCE

- Match device to the loss.
- Aids need speech recognition and dynamic range.
- Bone devices bypass the canal and middle ear.
- Implant for severe-profound with poor speech.
- Brainstem implant only if there's no nerve.
- Bone-conduction devices: skin-drive, percutaneous or active transcutaneous; ABI when the cochlear nerve cannot be used.

1 CHAPTER ONE Overview

The right device for the loss.

WHICH DEVICE?

SITUATION	CONSIDER
Sensorineural loss with useful speech understanding	Air-conduction hearing aid
Chronic discharge, canal atresia	Bone-anchored device
Single-sided deafness	CROS aid, bone-anchored device, cochlear implant
Moderate to severe loss, cannot tolerate an aid	Middle ear implant
Severe to profound loss with poor speech understanding	Cochlear implant

SITUATION	CONSIDER
No cochlear nerve, NF2	Auditory brainstem implant

1.1 Beyond devices

Assistive listening devices (loop systems, FM and remote microphones), alerting devices, speech reading, auditory training and speech conservation.

1.2 Two ears are better than one

Bilateral fitting gives the *head shadow* effect, *binaural summation* (sound seems about 3 dB louder) and *squelch* (better hearing in noise), and restores sound localisation.

2 CHAPTER TWO Hearing aids

Amplification.

2.1 Components

A microphone, amplifier and receiver (speaker), powered by a battery, with digital processing that shapes gain by frequency and compresses loud sounds.

2.2 Styles

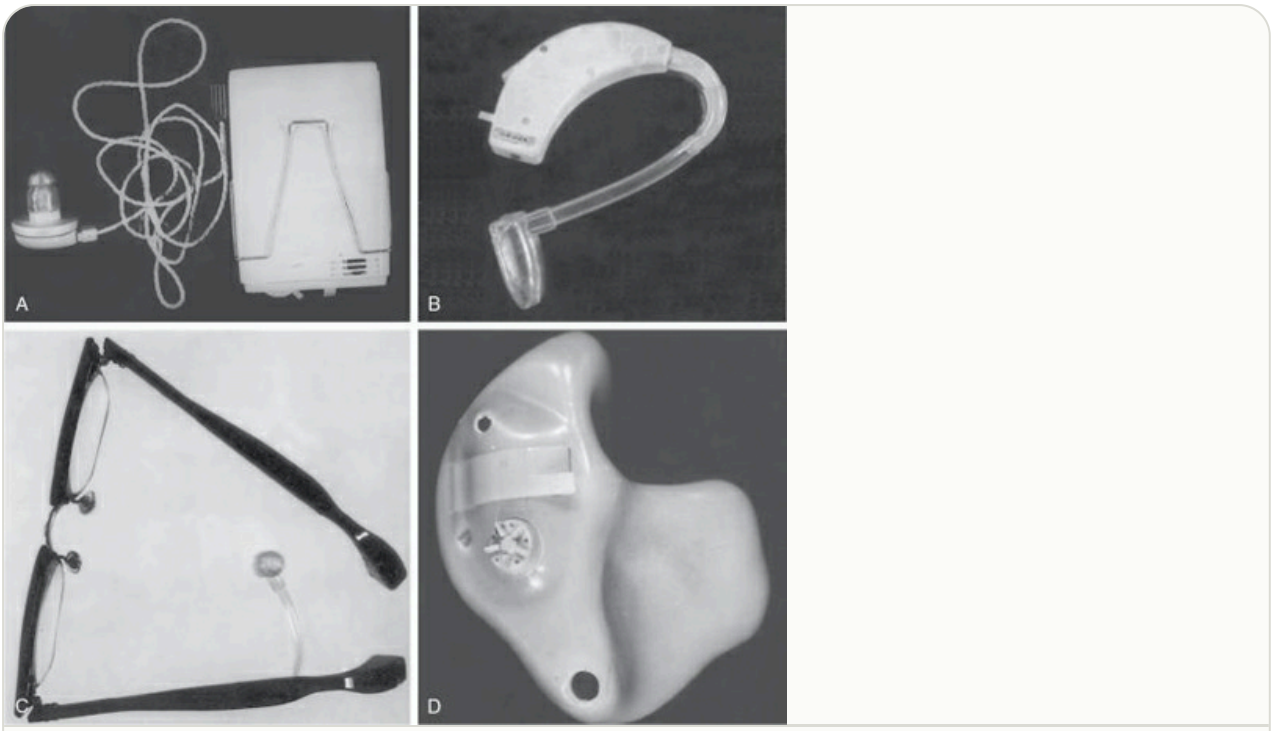


FIGURE 1 Types of hearing aid: (A) body-worn, (B) behind-the-ear, (C) spectacle and (D) in-the-ear.

Dhingra, fig. 20.1

- **Body-worn:** high power; now rare.
- **Behind-the-ear:** the most versatile, from mild to profound loss, and the choice for children because the mould can be changed as the ear grows.
- **Receiver-in-canal and open fit:** for high-frequency loss with good low tones; avoid the occlusion effect.
- **In-the-ear, in-the-canal, completely-in-canal:** cosmetic, for mild to moderate loss in adults with good dexterity.

2.3 Who does well

Speech discrimination of at least about 50%, and a wide *dynamic range* between the threshold and the level of discomfort.

2.4 Problems

2.4.1 Recruitment

Abnormal growth of loudness in a cochlear loss narrows the dynamic range: soft sounds are missed but loud sounds are intolerable. Compression circuits help.

2.4.2 Occlusion effect

A blocked canal makes the patient's own voice boom. Venting reduces it, but venting also loses low-frequency gain.

2.4.3 Feedback

Whistling from sound leaking back to the microphone; better moulds and feedback cancellation help.

CHAPTER THREE

3 Bone-conduction devices

Around the canal and middle ear.

3.1 How they work

Sound vibrates the skull directly and reaches both cochleas, bypassing the canal and middle ear.

3.2 Types

Conventional bone-conduction aids on a headband or spectacles; *percutaneous* bone-anchored devices fixed to an osseointegrated titanium implant through the skin; and *transcutaneous* (magnetic) devices under intact skin.



FIGURE 2 First, a bone-anchored device: titanium fixture, abutment and sound processor. Second, the device worn.
Dhingra, figs 20.2 and 20.3

3.3 Indications

- Chronic discharging ears that cannot tolerate an aid.
- Canal atresia and microtia.
- Conductive or mixed loss where surgery is unsuitable.
- Single-sided deafness with a good other ear (about 20 dB or better), sending sound across the skull.

3.4 Practical points

Osseointegration takes about three months, longer in children. Complications include failed integration and skin reactions around the abutment. A bone-anchored device does not restore localisation.

WHERE THE BOOKS DIFFER · SINGLE-SIDED DEAFNESS

Dhingra	Lee
Bone-anchored devices have replaced CROS aids.	CROS aids remain an option.

3.5 CROS and BiCROS aids

A microphone on the deaf side sends sound wirelessly to the better ear; BiCROS adds amplification for a better ear that also has some loss.

4 CHAPTER FOUR Middle ear implants

Driving the ossicles directly.

A transducer fixed to an ossicle or the round window, driven electromagnetically or piezoelectrically. For adults with moderate to severe sensorineural or mixed loss who are unhappy with conventional aids, and some conductive losses. The canal is left open.

5 CHAPTER FIVE Cochlear implants

Bypassing the cochlea.

5.1 Principle

An external microphone and processor convert sound into electrical patterns, sent across the skin to an internal receiver, which stimulates the spiral ganglion cells through an electrode array in the scala tympani. The cochlea's place coding is reproduced: basal electrodes for high pitches, apical for low.

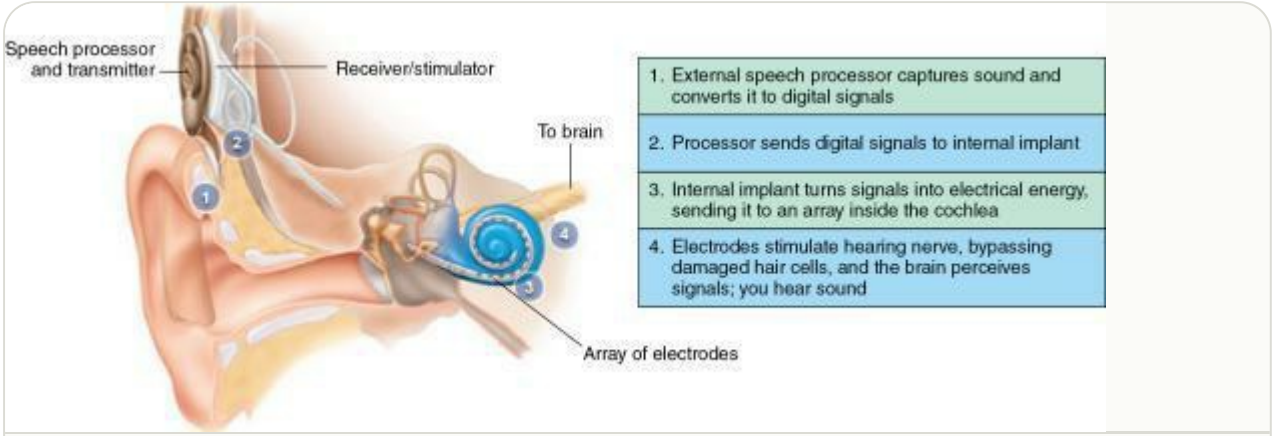


FIGURE 3 Principle of the cochlear implant.
Dhingra, fig. 20.9

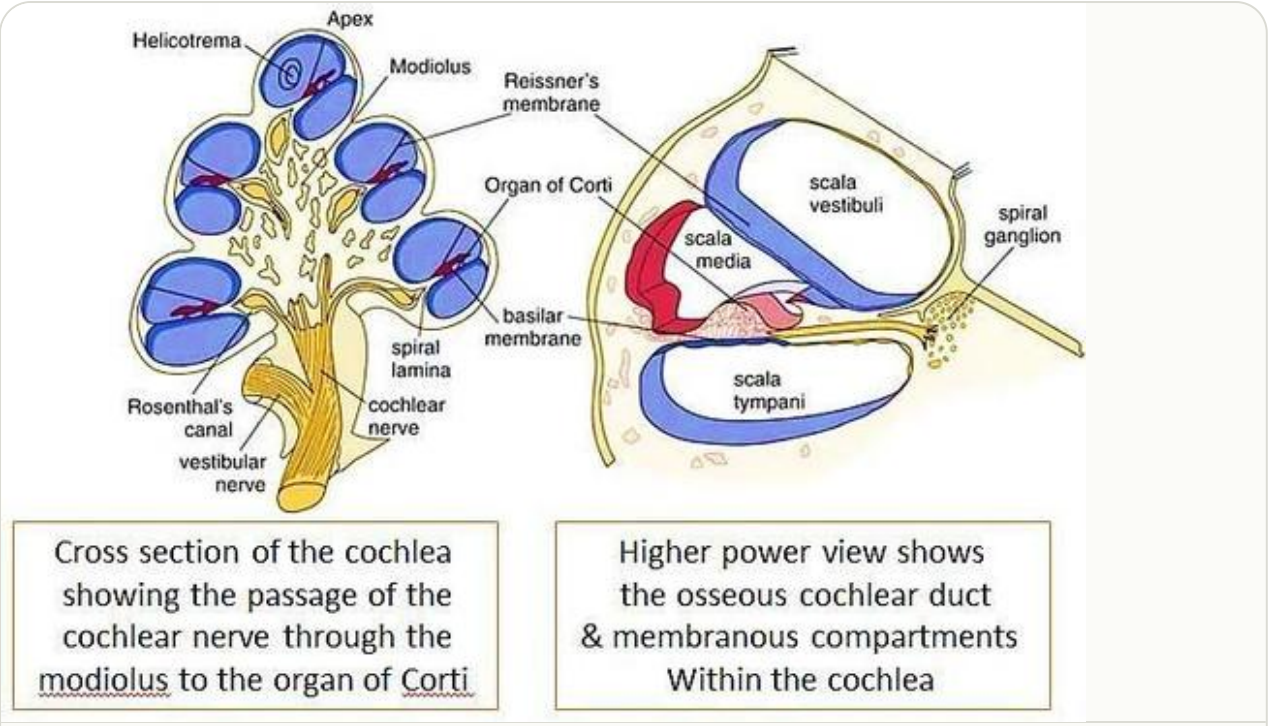


FIGURE 4 Cross-section of the cochlea: the electrode lies in the scala tympani, next to the spiral ganglion.

Anatomy atlas · ENT Steps p. 69

5.2 Candidates

5.2.1 Adults

Severe to profound bilateral sensorineural loss with limited benefit from well-fitted aids. Lee gives moderate to profound loss with an aided sentence score of 50% or less in the ear to be implanted and 60% or less best-aided.

5.2.2 Children

From about twelve months, with severe to profound loss and little benefit after a three to six month trial of aids. The trial is waived after meningitis, when ossification threatens.

WHERE THE BOOKS DIFFER · AGE OF IMPLANTATION

Dhingra	Lee
From twelve months.	Earlier implantation is safe and may help language.

5.2.3 Outcomes

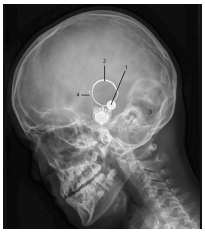
> Duration of deafness and pre-operative speech scores explain most of the variation. Postlingually deafened adults often achieve high sentence scores; children implanted early commonly join mainstream schools; prelingually deaf adults gain mostly sound awareness, because the window for language (birth to about two years) has passed.

5.3 Assessment

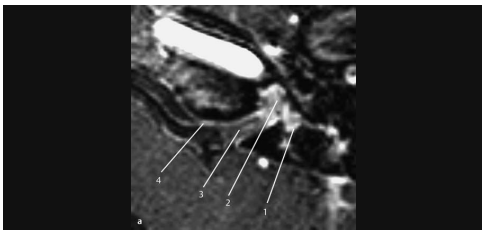
Medical review; audiology with an aid trial; MRI to show the cochlear nerve and a fluid-filled cochlea; HRCT for bony anatomy, malformations, a narrow internal auditory canal (under 3 mm) and surgical hazards; psychological and speech assessment.

IMAGING IN THE APP 4 cases

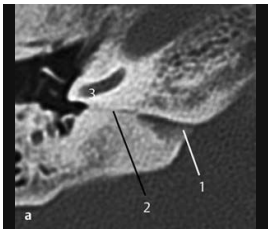
A cochlear implant in place, labyrinthitis ossificans, a gusher ear and iatrogenic fenestration.



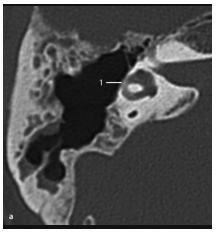
Cochlear Implantation
After surgery · 8 images



Labyrinthitis Ossificans (Ossification after Meningitis)
Bone disease · 14 images



Gusher
Infection & inflammation · 3 images



Iatrogenic Fenestration
After surgery · 2 images

5.4 Contraindications

Complete cochlear or labyrinthine aplasia (Michel), complete ossification, and an absent cochlear nerve. Retrocochlear disease is a relative contraindication.

5.5 The operation

Cortical mastoidectomy, posterior tympanotomy through the facial recess, entry into the scala tympani through the round window or a cochleostomy, slow insertion

of the array, sealing, and testing of the device. *Soft surgery* (a flexible array, the round window route, slow insertion, no suction of perilymph, and steroids) preserves residual hearing. In an ossified cochlea, the basal turn is drilled out or a split array used.

WHERE THE BOOKS DIFFER · ROUTE INTO THE COCHLEA

Dhingra	Lee
Describes a cochleostomy.	Prefers the round window.

OPERATIVE GUIDES IN THEATRE 5 operations

Bone-Anchored Hearing Devices
Bone-anchored devices

Implantable Middle Ear Devices
Implantable middle ear devices

Cochlear Implantation in Adults
Adult implant

Pediatric Cochlear Implantation
Paediatric implant

Ossicular Chain Reconstruction
Ossiculoplasty

5.6 After surgery

Activation at about two to four weeks, then repeated *mapping* of each electrode and intensive auditory therapy; sound quality improves over three to six months.

5.7 Vaccination

Implant recipients are vaccinated against pneumococcus (PCV13, then PPSV23 at least eight weeks later), because meningitis is commoner in implanted children, especially with malformations or a CSF leak.

5.8 Complications

- Flap breakdown and device extrusion.
- Device failure, about 4% over a lifetime; revision through the same track.
- Facial nerve injury (under 1%) and facial stimulation, managed by switching off

electrodes.

- Taste change, vertigo, tinnitus.
- CSF gusher, especially with malformations.
- Meningitis.

PEARLS

- Bilateral implants restore head shadow, summation and squelch, and localisation.
- Inner ear malformations raise the risk of an anomalous facial nerve, gusher and meningitis.
- Auditory neuropathy can do well with an implant.
- Acute otitis media in an implanted child is treated promptly, intravenously if severe.

CHAPTER SIX

6 Auditory brainstem implant

When there is no nerve.

An electrode placed on the cochlear nucleus, through the lateral recess of the fourth ventricle, for patients without a usable cochlear nerve: neurofibromatosis type 2, nerve aplasia, or complete cochlear ossification. Results are more modest than with cochlear implants.

THE IMPLANT FAMILY TODAY (OXFORD TEXTBOOK)

Implantable devices act at different levels of the pathway. **Bone-conduction hearing devices** bypass the outer and middle ear. *Skin-drive* (passive transcutaneous) devices, such as Baha Attract and the adhesive Adhear, lose energy through the skin and are being superseded. *Direct-drive* devices are either *percutaneous*, with an abutment through the skin (Baha Connect, Ponto), or *active transcutaneous*, with the vibrator implanted under intact skin (Bonebridge, Osia). UK criteria require a conductive or mixed loss, or single-sided deafness, with bone conduction in the fitting range and stable (no more than 15 dB change at two or more frequencies over two years), failure or unsuitability of a conventional or CROS aid after at least four weeks, and benefit from a trial on a headband for at least 14 days. **Active middle ear implants**, most often the Vibrant Soundbridge, drive the incus, stapes or round window with a floating mass transducer, and suit patients who cannot wear an aid but have a cochlea worth amplifying.

Cochlear implantation now aims to preserve the cochlea and any residual hearing: a round window or cochleostomy approach into the scala tympani, slow insertion over minutes to avoid basilar membrane rupture, and steroids. Children with profound bilateral loss are implanted on both sides, simultaneously, ideally between 9 and 12 months. In aural atresia, a headband bone-conduction aid is started by about six months, with a definitive implant at about four to five years. **Auditory brainstem implants** stimulate the cochlear nucleus when a cochlear implant cannot work: neurofibromatosis type 2, cochlear or nerve aplasia, and severe cochlear ossification after meningitis or fracture. In adults, treating hearing loss may slow cognitive decline.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Hearing aid types** and who benefits.
- **Recruitment** and dynamic range.
- **Bone-anchored devices:** indications and osseointegration.
- **Cochlear implant:** principle, candidacy, assessment, contraindications, operation, complications.
- **Auditory brainstem implant:** indications.

7.2 Common pitfalls

COMMON PITFALLS

- Fitting an aid to an asymmetric loss without excluding a retrocochlear cause.
- Delaying implantation after meningitis.
- Forgetting pneumococcal vaccination.

Written for this app from Oxford Textbook of Otolaryngology ch. 16, Oxford Specialist Handbook ch. 14, Dhingra ch. 20 and K. J. Lee ch. 19, 20. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

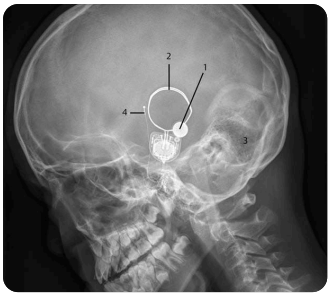
IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

INNER EAR & IAC · AFTER SURGERY

Cochlear Implantation



Dysfunction of the cochlear implant after head trauma. Plain film, lateral skull view. This view demonstrates dislocation of the magnet (1) from its…

INNER EAR & IAC · BONE DISEASE

Labyrinthitis Ossificans (Ossification after Meningitis)

IMAGING FINDINGS

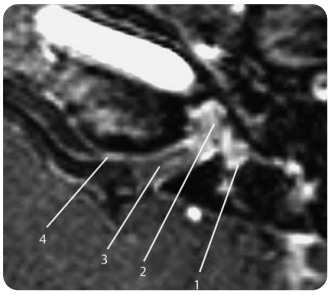
- Loss of the bright fluid signal of the cochlea on high-resolution T2
- Later: bone in the basal turn and canals on CT
- Early: enhancement of the labyrinth

TOP DIFFERENTIALS

- Labyrinthine or cochlear aplasia
- Cochlear otosclerosis
- Intralabyrinthine schwannoma

CHECKLIST

- After meningitis: urgent MRI and cochlear implantation
- Report patency of the basal turn



MRI, T1-weighted with gadolinium enhancement, axial. There is enhancement of the vestibule (1), cochlear contents (2), and the…



Axial bone CT in a 2-year-old boy with bilateral sensorineural hearing loss

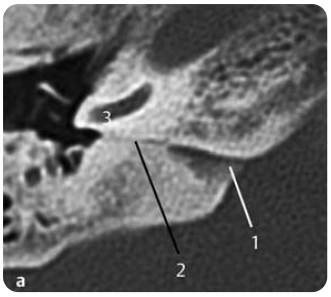
Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 1116

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1116.

INNER EAR & IAC · INFECTION & INFLAMMATION

Gusher

- A wide communication between the subarachnoid space and the perilymph (a defect at the fundus of the IAC or a dilated cochlear aqueduct) lets CSF pour out when the footplate is opened.
- CT clues: a bulbous, widened lateral IAC with a deficient fundus, absent modiolus and a corkscrew cochlea (X-linked incomplete partition type III, POU3F4).
- In a boy with a congenital mixed or conductive loss, review the CT for these signs before any stapes surgery.

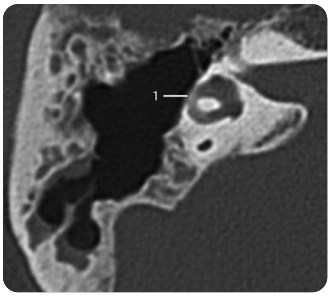


CT, axial. At stapedotomy, abundant flow of inner ear fluid, the gusher phenomenon, made further surgical exploration impossible and the hole i…

MIDDLE EAR & MASTOID · AFTER SURGERY

Iatrogenic Fenestration

- In fenestrated patients, no actual compression due to masses will be seen.
- Beware of suction of the cavity for cleaning. This must be performed very carefully to prevent severe vertigo and inner ear dysfunction.



CT, axial. Before the availability of microsurgical techniques and the use of ossicular replacement prostheses, mobilization of the footplate was…